

Functions: Composite, Inverse & Logarithmic



Domain and range

- 1 Find the domain of $f(x) = \frac{1}{x^2 - 9}$.
 - 2 Find the domain of $g(x) = \sqrt{7 - 2x}$.
 - 3 Find the range of $f(x) = -(x - 2)^2 + 5$.
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Composite functions

- 4 Given $f(x) = x^2 + 1$ and $g(x) = 2x - 3$, find $(f \circ g)(x)$.
 - 5 Given $f(x) = \sqrt{x + 4}$ and $g(x) = x^2 - 4$, find $(f \circ g)(x)$.
 - 6 Using the same f and g as the previous question, find $(g \circ f)(x)$ and state its domain.
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Inverse functions

- 7 Find $f^{-1}(x)$ for $f(x) = \frac{2x - 5}{3}$.
 - 8 Find $f^{-1}(x)$ for $f(x) = x^3 + 2$.
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Verifying inverses

- 9 Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x - 1}{5}$ are inverses by showing $f(g(x)) = x$.
 - 10 Does $f(x) = x^2$ (for all real x) have an inverse function? Explain.
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Logarithm laws

- 11 Expand $\log_b \left(\frac{x^3 y}{z^2} \right)$ using the laws of logarithms.
- 12 Condense $2\log_5 x + 3\log_5 y - \log_5 z$ into a single logarithm.
- 13 Evaluate $\log_2 32 - \log_2 4$ without a calculator.

Solving logarithmic equations

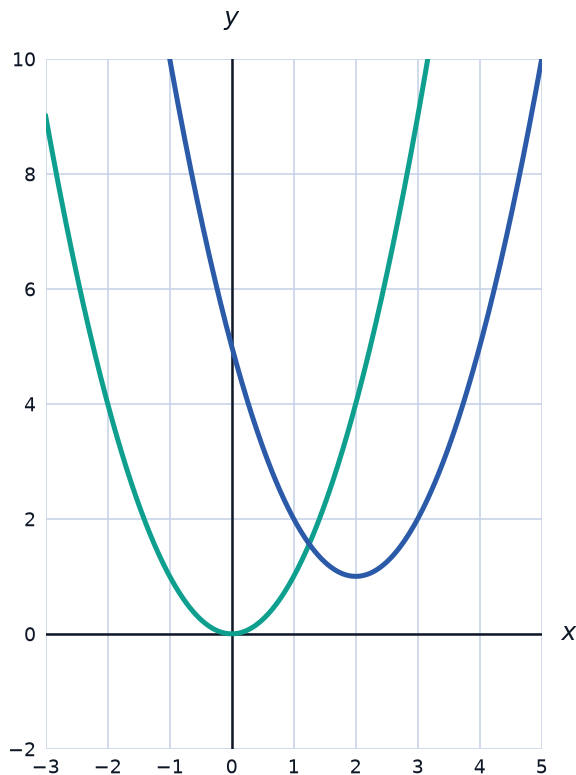
- 14 Solve $\log_4(x^2) = 3$.
- 15 Solve $\ln x + \ln(x - 3) = \ln 10$.

Exponential equations and logs as inverses

- 16 Solve $e^{2x-1} = 20$, giving your answer to three decimal places.
- 17 The functions $f(x) = \log_3 x$ and $g(x) = 3^x$ are inverses. Use this fact to solve $3^x = 81$ without a calculator, and state the value of $g(f(16))$.

Transformations of functions

- 18 The graph shows $f(x) = x^2$ (teal) and $g(x) = (x - 2)^2 + 1$ (indigo). Describe the transformation that maps f onto g .



- 19 If $h(x) = -f(x + 3)$ where $f(x) = x^2$, describe the transformations applied to f to obtain h .

A well-designed word problem: investment growth and doubling time

- 20** This question has three parts. An investment fund grows according to $A(t) = A_0 e^{0.045t}$, where A_0 is the initial amount (in dollars) and t is time in years. (a) If $A_0 = 8000$ dollars, find $A(10)$ to the nearest dollar. (b) Find the time t , to the nearest tenth of a year, for the investment to double. (c) Solve for t in terms of A and A_0 — that is, find the inverse function $t(A)$.
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Multiple choice — easy

- 21** If $f(x) = -4x + 7$ and $g(x) = -2x - 5$, find $(f \circ g)(7)$.
- a. 83
 - b. 85
 - c. -19
 - d. 81
- 22** If $f(x) = -x + 2$ and $g(x) = -4x + 3$, find $(f \circ g)(-1)$.
- a. -5
 - b. 7
 - c. -3
 - d. -7
- 23** If $f(x) = x$ and $g(x) = -3x + 3$, find $(f \circ g)(6)$.
- a. -13
 - b. -15
 - c. -12
 - d. -17

24 If $f(x) = x + 7$ and $g(x) = -3x + 9$, find $(f \circ g)(6)$.

a. 0

b. -9

c. -4

d. -2

25 If $f(x) = 2x - 6$ and $g(x) = -3x + 1$, find $(f \circ g)(3)$.

a. -20

b. -22

c. -8

d. -24

26 If $f(x) = -3x + 8$ and $g(x) = -4x + 7$, find $(f \circ g)(8)$.

a. 85

b. 81

c. -25

d. 83

27 If $f(x) = x$ and $g(x) = 3x - 9$, find $(f \circ g)(4)$.

a. 6

b. 5

c. 1

d. 3

28 If $f(x) = 2x + 1$ and $g(x) = -2x - 7$, find $(f \circ g)(4)$.

a. -15

b. -29

c. -31

d. -27

29 If $f(x) = 2x + 7$ and $g(x) = 4x + 4$, find $(f \circ g)(5)$.

a. 57

b. 53

c. 24

d. 55

30 If $f(x) = x - 1$ and $g(x) = 2x + 4$, find $(f \circ g)(-4)$.

a. -3

b. -7

c. -4

d. -5

31 If $f(x) = x - 8$ and $g(x) = -4x - 8$, find $(f \circ g)(1)$.

a. -12

b. -18

c. -22

d. -20

- 32 If $f(x) = 4x + 8$ and $g(x) = 2x - 2$, find $(f \circ g)(-7)$.
- a. -56
 - b. -16
 - c. -58
 - d. -54
- 33 If $f(x) = 4x - 6$ and $g(x) = -2x - 6$, find $(f \circ g)(-5)$.
- a. 8
 - b. 12
 - c. 10
 - d. 4
- 34 If $f(x) = -4x + 5$ and $g(x) = 2x + 3$, find $(f \circ g)(-2)$.
- a. 11
 - b. 7
 - c. -1
 - d. 9
- 35 If $f(x) = 2x - 3$ and $g(x) = -4x + 2$, find $(f \circ g)(6)$.
- a. -49
 - b. -45
 - c. -22
 - d. -47

Multiple choice — medium

36 Find $f^{-1}(x)$ for $f(x) = \frac{6x - 5}{8}$.

a. $\frac{6x + 5}{8}$

b. $\frac{8x + 5}{6}$

c. $\frac{8x - 5}{6}$

d. $\frac{6x - 5}{8}$

37 Find $f^{-1}(x)$ for $f(x) = \frac{2x - 16}{5}$.

a. $\frac{2x - 16}{5}$

b. $\frac{5x - 16}{2}$

c. $\frac{5x + 16}{2}$

d. $\frac{2x + 16}{5}$

38 Find $f^{-1}(x)$ for $f(x) = \frac{8x - 1}{5}$.

a. $\frac{5x - 1}{8}$

b. $\frac{8x + 1}{5}$

c. $\frac{5x + 1}{8}$

d. $\frac{8x - 1}{5}$

39 Find $f^{-1}(x)$ for $f(x) = \frac{9x - 13}{4}$.

a. $\frac{9x - 13}{4}$

b. $\frac{4x - 13}{9}$

c. $\frac{9x + 13}{4}$

d. $\frac{4x + 13}{9}$

40 Find $f^{-1}(x)$ for $f(x) = \frac{3x - 18}{2}$.

a. $\frac{2x + 18}{3}$

b. $\frac{3x - 18}{2}$

c. $\frac{2x - 18}{3}$

d. $\frac{3x + 18}{2}$

41 Find $f^{-1}(x)$ for $f(x) = \frac{5x - 14}{3}$.

a. $\frac{5x - 14}{3}$

b. $\frac{3x - 14}{5}$

c. $\frac{3x + 14}{5}$

d. $\frac{5x + 14}{3}$

42 Find $f^{-1}(x)$ for $f(x) = \frac{2x - 10}{7}$.

a. $\frac{7x + 10}{2}$

b. $\frac{2x - 10}{7}$

c. $\frac{7x - 10}{2}$

d. $\frac{2x + 10}{7}$

43 Find $f^{-1}(x)$ for $f(x) = \frac{7x - 15}{4}$.

a. $\frac{4x + 15}{7}$

b. $\frac{7x - 15}{4}$

c. $\frac{7x + 15}{4}$

d. $\frac{4x - 15}{7}$

44 Find $f^{-1}(x)$ for $f(x) = \frac{6x - 4}{4}$.

a. $\frac{6x - 4}{4}$

b. $\frac{4x - 4}{6}$

c. $\frac{4x + 4}{6}$

d. $\frac{6x + 4}{4}$

45 Find $f^{-1}(x)$ for $f(x) = \frac{6x - 18}{4}$.

a. $\frac{4x + 18}{6}$

b. $\frac{4x - 18}{6}$

c. $\frac{6x - 18}{4}$

d. $\frac{6x + 18}{4}$

46 Find $f^{-1}(x)$ for $f(x) = \frac{5x - 2}{8}$.

a. $\frac{5x - 2}{8}$

b. $\frac{8x - 2}{5}$

c. $\frac{5x + 2}{8}$

d. $\frac{8x + 2}{5}$

47 Find $f^{-1}(x)$ for $f(x) = \frac{4x - 20}{2}$.

a. $\frac{2x - 20}{4}$

b. $\frac{2x + 20}{4}$

c. $\frac{4x - 20}{2}$

d. $\frac{4x + 20}{2}$

48 Find $f^{-1}(x)$ for $f(x) = \frac{2x - 11}{6}$.

a. $\frac{2x - 11}{6}$

b. $\frac{2x + 11}{6}$

c. $\frac{6x - 11}{2}$

d. $\frac{6x + 11}{2}$

49 Find $f^{-1}(x)$ for $f(x) = \frac{8x - 2}{5}$.

a. $\frac{5x + 2}{8}$

b. $\frac{8x + 2}{5}$

c. $\frac{8x - 2}{5}$

d. $\frac{5x - 2}{8}$

50 Find $f^{-1}(x)$ for $f(x) = \frac{6x - 3}{3}$.

a. $\frac{6x - 3}{3}$

b. $\frac{6x + 3}{3}$

c. $\frac{3x - 3}{6}$

d. $\frac{3x + 3}{6}$

Multiple choice — hard

51 Solve $\ln x + \ln(x - 4) = \ln 140$.

a. 14

b. 12

c. 16

d. -10

52 Solve $\ln x + \ln(x - 5) = \ln 6$.

a. 8

b. -1

c. 4

d. 6

53 Solve $\ln x + \ln(x - 4) = \ln 77$.

a. 11

b. -7

c. 9

d. 13

54 Solve $\ln x + \ln(x - 4) = \ln 221$.

a. 15

b. 17

c. 19

d. -13

55 Solve $\ln x + \ln(x - 6) = \ln 247$.

a. -13

b. 17

c. 21

d. 19

56 Solve $e^{3x-9} = 433$, giving your answer to three decimal places.

a. 5.524

b. 4.524

c. 6.024

d. 5.024

57 Solve $e^{3x-7} = 105$, giving your answer to three decimal places.

a. 3.385

b. 3.885

c. 4.885

d. 4.385

58 Solve $e^{4x-9} = 67$, giving your answer to three decimal places.

a. 2.801

b. 3.301

c. 3.801

d. 4.301

59 Solve $e^{4x-6} = 488$, giving your answer to three decimal places.

a. 4.048

b. 3.548

c. 3.048

d. 2.548

60 Solve $e^{2x-8} = 355$, giving your answer to three decimal places.

a. 6.436

b. 7.936

c. 6.936

d. 7.436

61 Evaluate $\log_3 648 - \log_3 8$ without a calculator.

a. 3

b. 5

c. 4

d. 640

62 Evaluate $\log_5 500 - \log_5 4$ without a calculator.

a. 496

b. 3

c. 4

d. 2

63 Evaluate $\log_3 5103 - \log_3 7$ without a calculator.

a. 7

b. 5

c. 5096

d. 6

64 Evaluate $\log_3 72 - \log_3 8$ without a calculator.

a. 64

b. 2

c. 3

d. 1

65 Evaluate $\log_2 32 - \log_2 2$ without a calculator.

a. 30

b. 3

c. 4

d. 5